

Fare engine acceptance: memo

Verdict: the engine fails acceptance. It priced 17 of the 38 journeys in the plan wrongly. Across the plan its prices come to +7.15 USD against the certified fares of the journeys that have a fare (+17.05 overcharged, -9.90 undercharged), and it returned a price for 7 journeys that have no fare on their medium at all.

What the vendor got wrong, by rule

Cash or medium not valid for a leg (no fare on the medium) (J04, J06, J08, J10, J31, J32, J37): the engine prices every leg on every medium. Under the feed, cash has no product at a subway fare gate (the cash rule for rapid transit is transfer only, so it can never start a journey), the mTicket app has no bus or subway product, and the CharlieCard and contactless products do not exist for commuter rail. Each of these journeys has no fare on its medium and the engine must say so.

Charlieticket priced as a local bus fare (J03): the only product a CharlieTicket holds for a local bus leg is the subway quick ticket at 2.40, not the 1.70 bus fare.

Transfer applied outside the two hour window or beyond the allowed count (J13, J16, J38): the bus to bus rule allows one transfer within two hours of the first departure, measured departure to departure; the subway to bus rule allows one; a third bus, or a bus boarded more than two hours after the first departure, is a new fare.

Transfer discount given to a medium or product that has none (J11, J15): cash carries no transfer rules between buses or onto the subway, and the quick ticket product has no transfer onto the subway from a bus.

Free silver line from the airport terminals charged (J20, J21): the SL1 leg from the airport terminals is a free fare product and its transfer to the Red Line at South Station is free as well.

Commuter rail interzone fare priced as a zone fare from boston (J27, J28): a journey between two stations outside Zone 1A is priced by the interzone product for the number of zones crossed, not by the zone fare from Boston.

Largest error and the window

The largest error in dollars is J28: certified 4.75, vendor 12.25, +7.50.

If the transfer window were cut to ninety minutes, exactly one journey's correct fare would change: J19, from 2.40 to 4.10, because its second leg departs 115 minutes after the first and so falls inside two hours but outside ninety minutes. Every other transfer in the plan is made within 45 minutes or already falls outside two hours.

Journey	Medium	Legs	Certified	Vendor	Match
J01	charliecard	1	1.70	1.70	Yes
J02	cash	1	1.70	1.70	Yes
J03	charlieticket	1	2.40	1.70	No
J04	mticket	1	no fare on this medium	1.70	No
J05	contactless_credit_debit	1	2.40	2.40	Yes
J06	cash	1	no fare on this medium	2.40	No
J07	cash	1	2.40	2.40	Yes
J08	cash	1	no fare on this medium	2.40	No
J09	charliecard	2	2.40	2.40	Yes
J10	cash	2	no fare on this medium	2.40	No
J11	charlieticket	2	4.80	2.40	No
J12	charliecard	2	2.40	2.40	Yes
J13	charliecard	2	4.10	2.40	No
J14	charliecard	2	1.70	1.70	Yes

Journey	Medium	Legs	Certified	Vendor	Match
J15	cash	2	3.40	1.70	No
J16	charliecard	3	3.40	1.70	No
J17	charliecard	3	2.40	2.40	Yes
J18	charliecard	2	2.40	2.40	Yes
J19	contactless_credit_debit	2	2.40	2.40	Yes
J20	cash	1	0.00	2.40	No
J21	charliecard	2	0.00	2.40	No
J22	charliecard	1	2.40	2.40	Yes
J23	charliecard	1	2.40	2.40	Yes
J24	charliecard	1	2.40	2.40	Yes
J25	cash	1	2.40	2.40	Yes
J26	mticket	1	8.00	8.00	Yes
J27	cash	1	3.25	8.00	No
J28	mticket	1	4.75	12.25	No
J29	mticket	1	2.40	2.40	Yes
J30	mticket	1	7.00	7.00	Yes
J31	charliecard	1	no fare on this medium	2.40	No
J32	contactless_credit_debit	1	no fare on this medium	2.40	No
J33	mticket	1	7.00	7.00	Yes
J34	cash	1	9.75	9.75	Yes
J35	cash	1	2.40	2.40	Yes
J36	contactless_credit_debit	1	6.50	6.50	Yes
J37	mticket	2	no fare on this medium	12.15	No
J38	contactless_credit_debit	3	4.10	2.40	No